

AMENDMENT NO. 1 MARCH 2022

TO

IS 1893 (PART 2) : 2014 CRITERIA FOR EARTHQUAKE RESISTANT DESIGN OF STRUCTURES

PART 2 LIQUID RETAINING TANKS

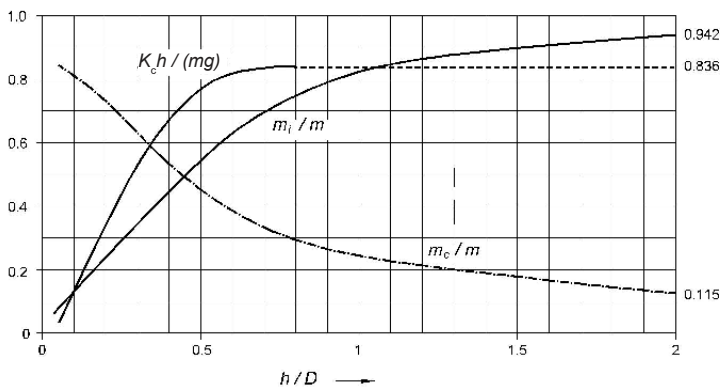
(Fifth Revision)

(Page 1, clause 3.0, line 15) — Substitute ' $\bar{h}$ ' for ' h '.

(Page 3, clause 4.2.1.2) — Insert the following at the end:

‘Expressions for these parameters are given in Annex A’.

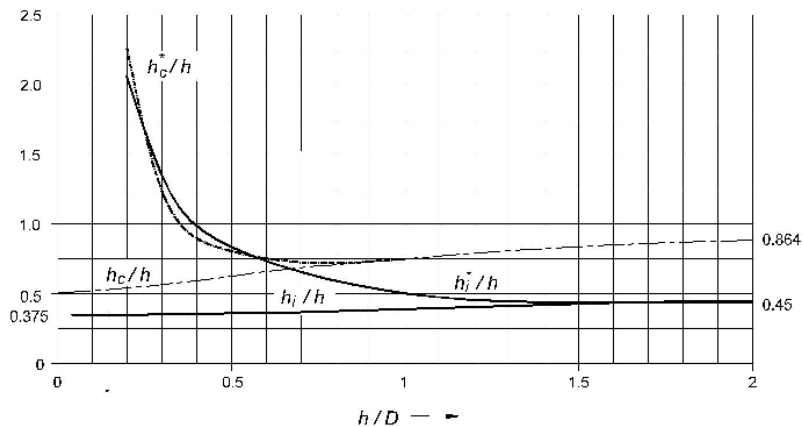
[Page 3, Fig. 2(a) and 2(b)] — Substitute the following figures for the existing:



2(a) IMPULSIVE AND CONVECTIVE MASS AND CONVECTIVE SPRING STIFFNESS

Price Group 4

[Page 3, Fig 2 (b)] — *Substitute the following for the existing:*



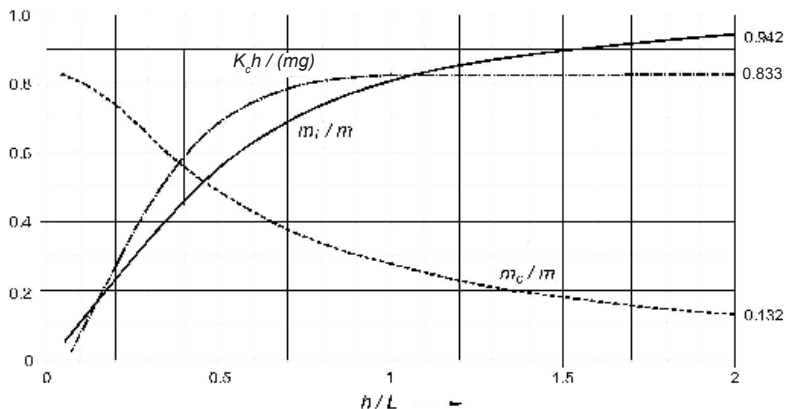
2(b) HEIGHTS OF IMPULSIVE AND CONVECTIVE MASSES

FIG. 2 PARAMETERS OF THE SPRING MASS MODEL FOR CIRCULAR TANK

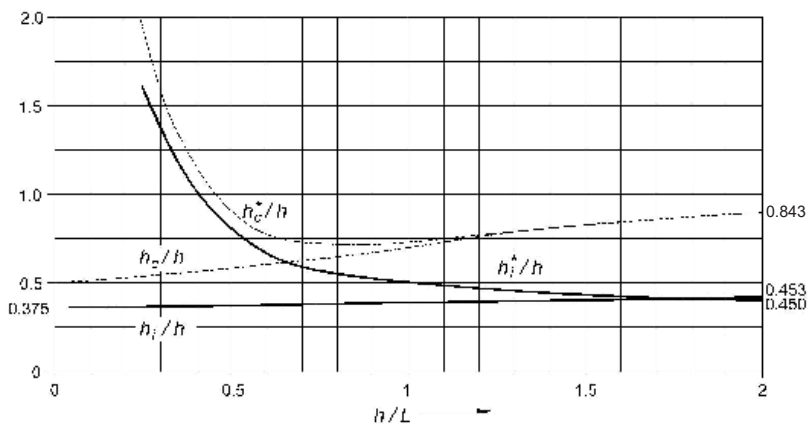
(Page 3, clause 4.2.3, line 5) — Insert ‘,’ after ‘ h_c ’.

(Page 4, clause 4.3.1.2, line 7) — Substitute ‘ $\bar{h}$ ’ for ‘ h ’.

[Page 4, Fig. 3(a) and 3(b)] — Substitute the following figures for the existing:



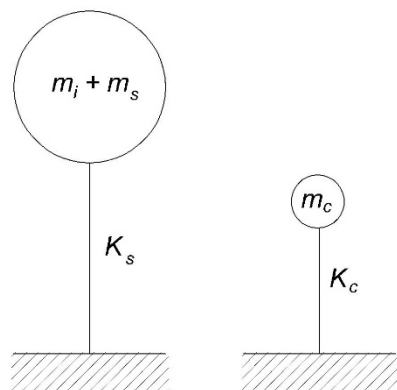
3(a) IMPULSIVE AND CONVECTIVE MASS AND CONVECTIVE SPRING STIFFNESS



3(b) HEIGHTS OF IMPULSIVE AND CONVECTIVE MASSES

FIG. 3 PARAMETERS OF THE SPRING MASS MODEL FOR RECTANGULAR TANK

[Page 5, Fig. 4(d)] — Substitute the following figure for the existing:



4(d) EQUIVALENT UNCOUPLED SYSTEM TANK

(Page 5, Fig. 5) — Substitute the following figure for the existing:

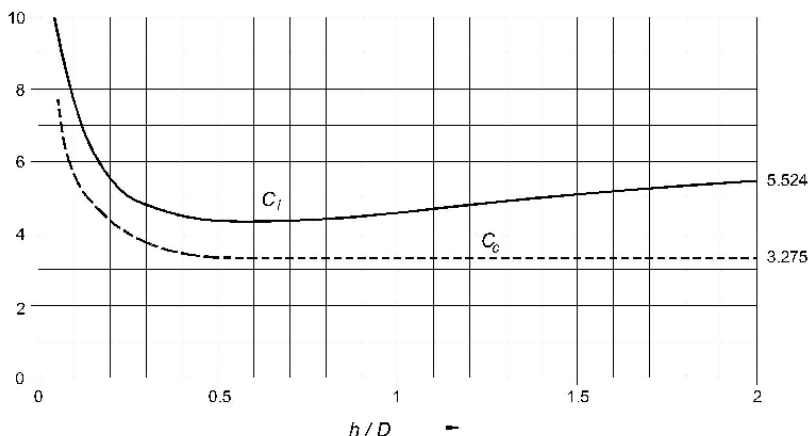


FIG. 5 COEFFICIENT OF IMPULSIVE (C_i) AND CONVECTIVE (C_c) MODE TIME PERIOD FOR CIRCULAR TANK

Amendment No. 1 to IS 1893 (Part 2) : 2014

(Page 6, clause 4.3.2.1, equation for T_1) — Substitute ' T_c ' for ' T_1 '.

[Page 6, clause 4.3.2.2(a), equation for T_c] — Substitute ' g ' for ' ρ '.

[Page 6, clause 4.3.2.2(b), line 4] — Substitute ' g ' for ' ρ '.

(Page 7, Table 2) — Substitute the following table for the existing:

Table 2 Suggested Values of ' R ' for Elevated Tanks

(Clause 4.5)

Sl No.	Type of Elevated Tank	Value of ' R '
(1)	(2)	(3)
i)	Tank supported on masonry shaft (Not permitted in zones III, IV and V):	
	a) Shaft reinforced with horizontal bands	1.5
	b) Shaft reinforced with horizontal bands and vertical bars	2.0
ii)	Tank supported on RCC shaft with :	
	a) Reinforcement in two curtains (in both directions)	2.5
	b) Reinforcement in two curtains (in both directions) and with stiffened openings and bracings	3.0
iii)	Tank supported on RCC framed staging ¹⁾	
	a) Ordinary moment resisting frame (OMRF) type ²⁾	2.0
	b) Special moment resisting frame (SMRF) conforming to ductility requirement of IS 13920	3.0
iv)	Tank supported on steel framed staging ¹⁾	
	a) Special moment resisting frame (SMRF) without diagonal brace	2.5
	b) Special moment resisting frame (SMRF) with diagonal brace	3.0

¹⁾ These R values are meant for liquid retaining tanks on frame type staging which are inverted pendulum type structures. These R values shall not be misunderstood for those given in other parts of IS 1893 for buildings and industrial frames.

²⁾ OMRF is permitted only in Zone II for tanks up to capacity 50 m³ and staging height of 12 m.

NOTE — P - Δ effect should be considered in the design of staging.

(Page 7, Table 3) — Substitute the following table for the existing:

Table 3 Suggested Values of ‘R’ for Ground Supported Tanks
(Clause 4.5)

Sl No.	Type of Ground Supported Tank	Value of ‘R’
(1)	(2)	(3)
i)	Masonry tank:	
	a) Masonry wall reinforced with horizontal bands (<i>Not permitted in zone IV and V</i>)	1.5
	b) Masonry wall reinforced with horizontal bands and vertical bars at corners and jams of openings	2.0
ii)	RCC or prestressed tank:	
	a) Fixed or hinged/pinned base tank (<i>see</i> Fig. 6a, 6b, 6c)	2.0
	b) Anchored flexible base tank (<i>see</i> Fig. 6d)	2.5
	c) Unanchored contained or uncontained tank (<i>see</i> Fig. 6e, 6f)	1.5
iii)	Steel tank:	
	a) Unanchored base	1.5
	b) Anchored base	2.5
iv)	Underground RC and steel tank (<i>see</i> Note)	3.0
NOTE — For partially buried tanks, value of <i>R</i> can be interpolated between ground supported and underground tanks based on depth of embedment.		

(Page 8, clause 4.6.3) — Insert the following clause at the end:

‘4.6.4 — For design, the horizontal impulsive coefficient (A_{hi}) shall not be less than that given in table below.’

Seismic zone	Ratio	In percent
II	0.015	1.5
III	0.020	2.0
IV	0.030	3.0
V	0.045	4.5

(Page 6, Fig. 7) — Substitute the following figure for the existing:

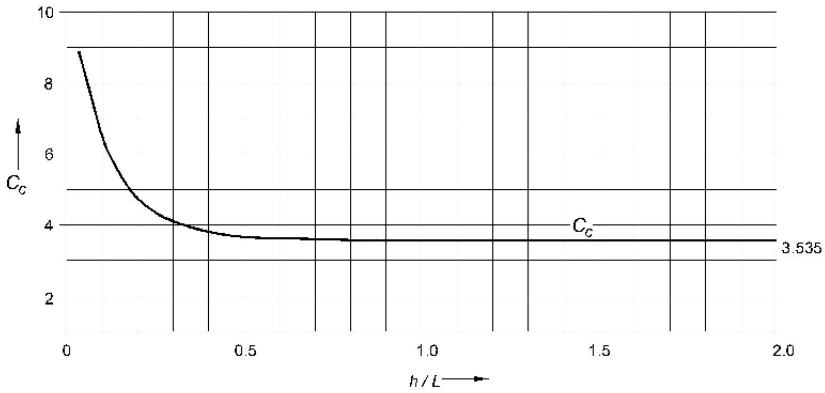


FIG. 7 COEFFICIENT OF CONVECTIVE MODE NATURAL PERIOD (C_c) FOR RECTANGULAR TANK

(Page 8, clause 4.7.2) — In the expression of ' M_c^* ', substitute ' h_c^* ' for ' h_{cg} '.

[Page 9, clause 4.9.1(a), equation for P_{ib}] — Substitute ' p_{ib} ' for ' P_{ib} ' and ' $(A_h)_i$ ' for ' $(A_h)_l$ '.

[Page 9, Fig. 8(a)] — Substitute " l " for ' l '.

[Page 10, clause 4.9.1(b)] — Substitute ' p_{ib} ' for ' P_{ib} '.

[Page 10, clause 4.9.1(b), equation for $Q_{ib}(x)$] — Substitute the following for the existing:

$$Q_{ib}(x) = 0.866 \frac{\sinh\left(1.732 \frac{x}{h}\right)}{\cosh\left(0.866 \frac{L}{h}\right)}$$

[Page 10, clause 4.9.2(a), equation for p_{cw}] — Substitute the following for the existing:

$$p_{cw} = Q_{cw}(y) (A_h)_c \rho g D \left(1 - \frac{1}{3} \cos^2 \phi \right) \cos \phi$$

[Page 10, clause 4.9.2(a), equation for $Q_{cb}(x)$] — Substitute ‘3.674’ for ‘0.3674’ in the expression for $Q_{cb}(x)$.

[Page 10, clause 4.9.2(a), equation for P_{cb}] — Substitute ‘ p_{cb} ’ for ‘ P_{cb} ’.

[Page 10, clause 4.9.2(b), equation for P_{cb}] — Substitute ‘ p_{cb} ’ for ‘ P_{cb} ’.

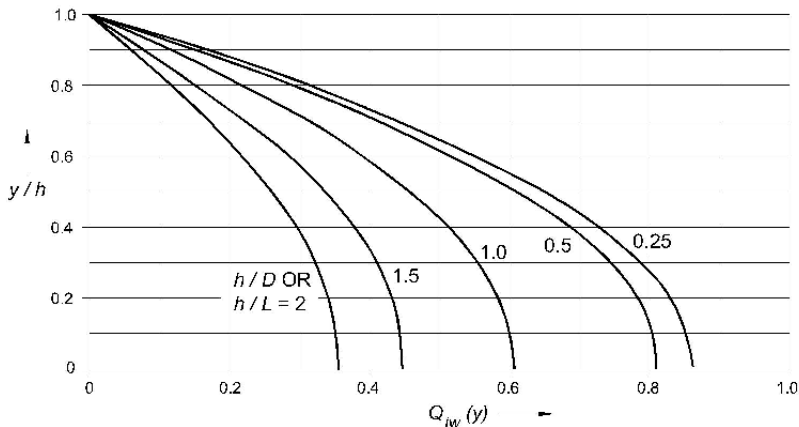
[Page 10, clause 4.9.2(b), equation for $Q_{cb}(x)$] — Substitute ‘1.25’ for ‘1.125’.

(Page 10, clause 4.9.5) — Substitute ‘ p_{ww} ’ for ‘ P_{ww} ’.

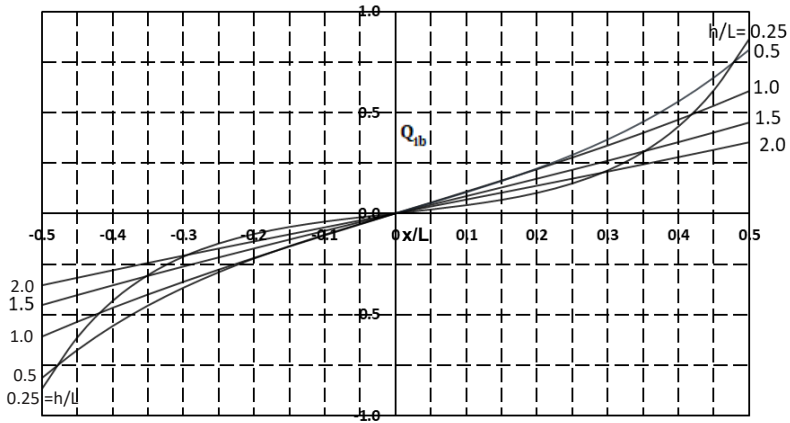
(Page 10, clause 4.10) — Substitute the following for the description of (S_a/g) in place of the existing:

‘ (S_a/g) = Average response acceleration coefficient to be taken as 2.5’.

[Page 11, Fig. 9(a) and 9(b)] — Substitute the following figures for the existing:



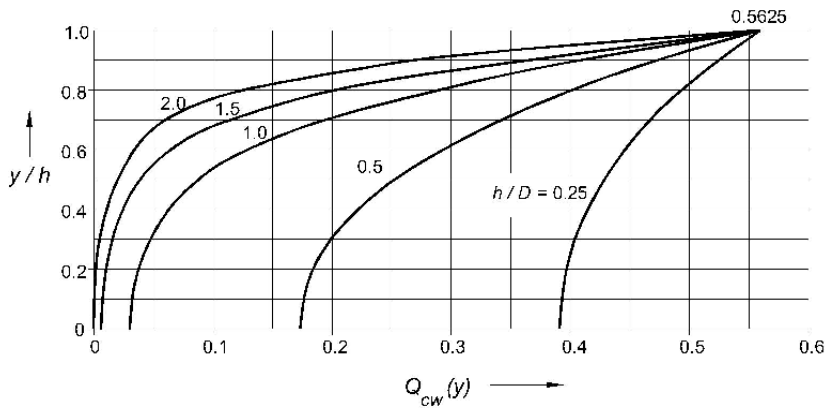
9(a) ON WALL OF CIRCULAR AND RECTANGULAR TANKS



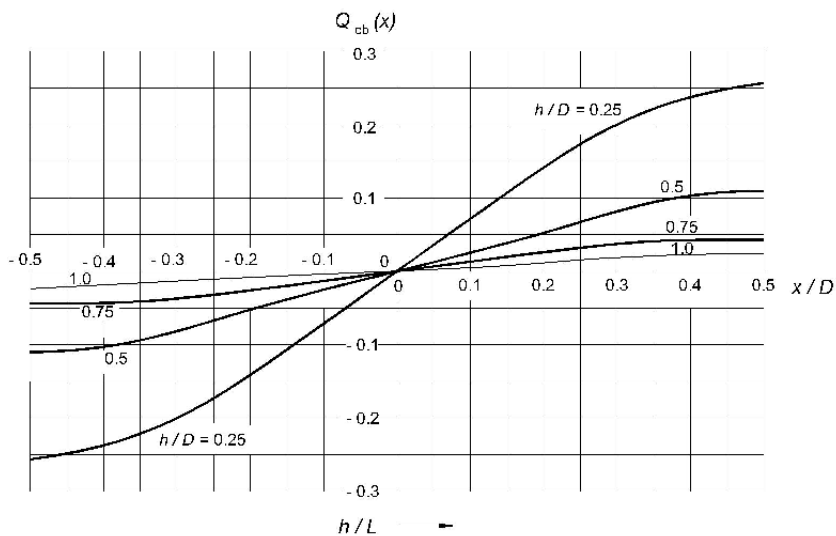
9(b) ON BASE OF RECTANGULAR TANK

FIG. 9 IMPULSIVE PRESSURE COEFFICIENT

[Page 12, Fig. 10(a) and 10(b)] — Substitute the following figures for the existing:



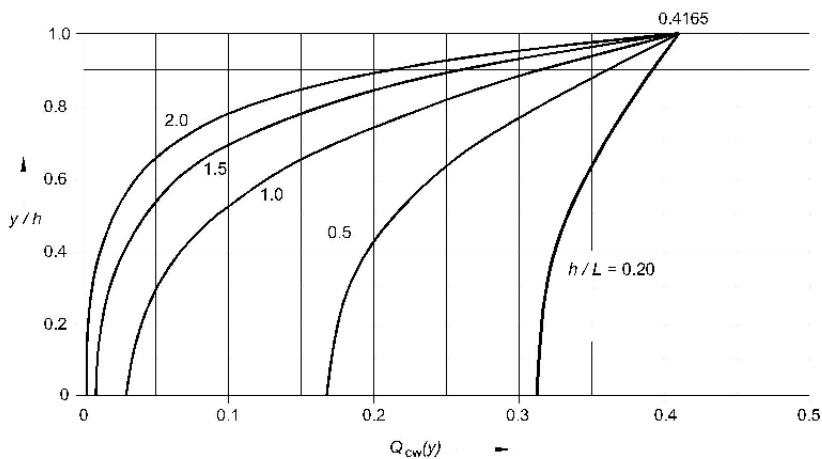
10(a) ON WALL



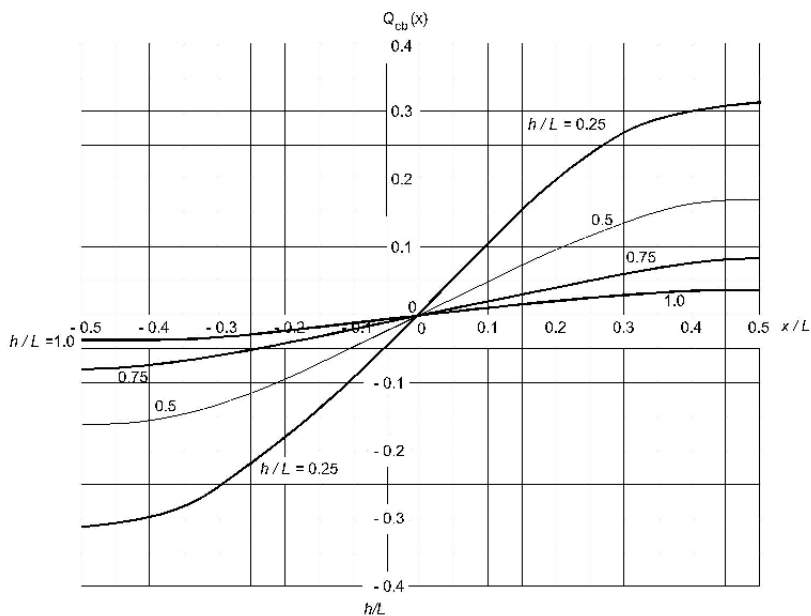
10(b) ON BASE

FIG. 10 CONVECTIVE PRESSURE COEFFICIENT FOR CIRCULAR TANK

[Page 13, Fig. 11(a) and 11(b)] — Substitute the following figures for the existing:



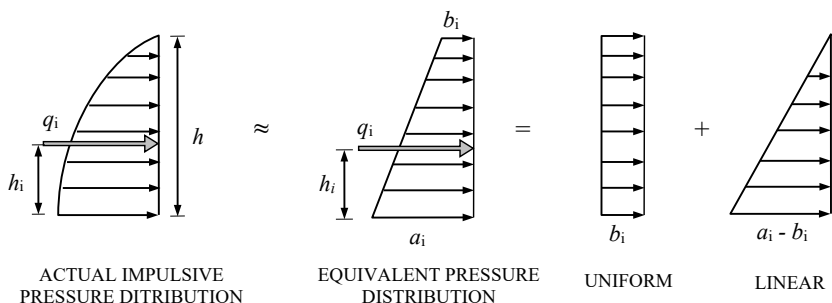
11(a) ON WALL



11(b) ON BASE

FIG. 11 CONVECTIVE PRESSURE COEFFICIENT FOR RECTANGULAR TANK

[Page 14, Fig. 12(b)] — Substitute the following figure for the existing:



12(b) EQUIVALENT LINEAR DISTRIBUTION ALONG WALL HEIGHT FOR IMPULSIVE PRESSURE

(Page 16, clause 6.5) — Insert the following at the end:

‘6.6 Cantilever Projections

6.6.1 Vertical Projections

Towers, tanks, parapets and other vertical cantilever projections attached to structures and projecting above the roof, shall be designed for five times the design horizontal seismic coefficient for impulsive mode (A_{hi}) as calculated in 4.5.

6.6.2 Horizontal Projections

All horizontal projections like walkways and balconies shall be designed for five times the design vertical seismic coefficient for impulsive mode (A_{vi}) as calculated in 4.10.1.

6.6.3 The increased design forces specified in 6.6.1 and 6.6.2 are only for designing the projecting parts and their connections with the main structures. For the design of the main structure, such increase shall not be considered.’

(Page 18, clause 8.5) — Insert the following Annex at the end:

ANNEX A

(Clause 4.2.1.2)

EXPRESSIONS FOR PARAMETERS OF SPRING MASS MODEL

For Circular Tank	For Rectangular Tank
$\frac{m_i}{m} = \frac{\tanh\left(0.866 \frac{D}{h}\right)}{0.866 \frac{D}{h}}$	$\frac{m_i}{m} = \frac{\tanh\left(0.866 \frac{L}{h}\right)}{0.866 \frac{L}{h}}$
$\frac{h_i}{h} = 0.375 \text{ for } h/D \leq 0.75$	$\frac{h_i}{h} = 0.375 \text{ for } h/L \leq 0.75$

For Circular Tank	For Rectangular Tank
$= 0.5 - \frac{0.09375}{h/D} \text{ for } h/D > 0.75$	$= 0.5 - \frac{0.09375}{h/L} \text{ for } h/L > 0.75$
$\frac{h_i^*}{h} = \frac{0.866 \frac{D}{h}}{2 \tanh\left(0.866 \frac{D}{h}\right)} - 0.125 \text{ for } h/D \leq 1.33$	$\frac{h_i^*}{h} = \frac{0.866 \frac{L}{h}}{2 \tanh\left(0.866 \frac{L}{h}\right)} - 0.125$
	for $h/L \leq 1.33$
$= 0.45 \text{ for } h/D > 1.33$	$= 0.45 \text{ for } h/L > 1.33$
$\frac{m_c}{m} = 0.23 \frac{\tanh\left(3.68 \frac{h}{D}\right)}{\frac{h}{D}}$	$\frac{m_c}{m} = 0.264 \frac{\tanh\left(3.16 \frac{h}{L}\right)}{\frac{h}{L}}$
$\frac{h_c}{h} = 1 - \frac{\cosh\left(3.68 \frac{h}{D}\right) - 1.0}{3.68 \frac{h}{D} \sinh\left(3.68 \frac{h}{D}\right)}$	$\frac{h_c}{h} = 1 - \frac{\cosh\left(3.16 \frac{h}{L}\right) - 1.0}{3.16 \frac{h}{L} \sinh\left(3.16 \frac{h}{L}\right)}$
$\frac{h_c^*}{h} = 1 - \frac{\cosh\left(3.68 \frac{h}{D}\right) - 2.01}{3.68 \frac{h}{D} \sinh\left(3.68 \frac{h}{D}\right)}$	$\frac{h_c^*}{h} = 1 - \frac{\cosh\left(3.16 \frac{h}{L}\right) - 2.01}{3.16 \frac{h}{L} \sinh\left(3.16 \frac{h}{L}\right)}$

For Circular Tank	For Rectangular Tank
$K_c = 0.836 \frac{mg}{h} \tanh^2 \left(3.68 \frac{h}{D} \right)$	$K_c = 0.833 \frac{mg}{h} \tanh^2 \left(3.16 \frac{h}{L} \right)$
$C_i = \left(\frac{1}{\sqrt{h/D} \left(0.46 - 0.3h/D + 0.067(h/D)^2 \right)} \right)$	—
$C_c = \frac{2\pi}{\sqrt{3.68 \tanh(3.68(h/D))}}$	$C_c = \frac{2\pi}{\sqrt{3.16 \tanh(3.16(h/L))}}$